The weak endpoint for derivatives of polynomial roots

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Abstract

Continuous roots of a $C^{n-1,1}$ polynomial curve have derivative in weak $L^{n/(n-1)}$, with a coefficient bound. The conclusion concerns exactly the catalogue target; no historical priority claim is made.

Catalogue scope. AIM 728; complete resolution. Canonical target at revision `1e56463`.

1 Statement and argument

The answer to the stated question is affirmative. The argument below combines known estimates for radicals and lower-degree polynomials with a maximal-function covering argument. In particular, it does not sum local weak-Lebesgue quasinorms, whose failure of additivity is the obstruction mentioned in [1, Open Problem 2].

For an interval I, write

$$\|f\|_{C^{m,1}(I)} = \max\left\{\max_{0\le s\le m}\|f^{(s)}\|_{L^\infty(I)}, \operatorname{Lip}_I(f^{(m)})\right\}, \qquad \|f\|_{p,\infty;I} = \sup_{u>0} u\,|\{|f|>u\}\cap I|^{1/p}.$$

Equivalent conventions for the first norm only change constants.

Theorem 1.1. *For every integer $n \ge 3$ there is a constant C_n with the following property. Let*

$$P_t(z) = z^n + \sum_{j=1}^{n} a_j(t) z^{n-j}, \qquad a_j \in C^{n-1,1}([0,1];\mathbb{C}),$$

and let $\lambda : (0,1) \to \mathbb{C}$ be any continuous root. Then, with $p = n/(n-1)$,

$$\|\lambda'\|_{p,\infty;(0,1)} \le C_n \max_{1\le j\le n} \|a_j\|^{1/j}_{C^{n-1,1}([0,1])}. \tag{728.1}$$

Consequently every root in the exact hypotheses of AIM 728 has derivative in $L^{n/(n-1),\infty}(I)$.

1.1 Established inputs and an elementary interpolation estimate

We use two previously established results, with their roles kept distinct. First, the complex radical estimate [1, Proposition 1 and Corollary 2], based on [2], says that if $a \in C^{j-1,1}(T;\mathbb{C})$, then every continuous jth root is absolutely continuous and

$$g_a(t) = \begin{cases} \dfrac{|a'(t)|}{j|a(t)|^{1-1/j}}, & a(t) \ne 0, \\ 0, & a(t) = 0 \end{cases} \quad \text{belongs to } L^{j/(j-1),\infty}(T). \tag{728.2}$$

*The solutions and partial results in this document were found by AI. The supplied research and the separate repository review are AI work; human peer review and formal verification are not claimed.

Its weak norm is bounded by a constant depending only on j times

$$\max\left\{\mathrm{Lip}_T(a^{(j-1)})^{1/j}|T|^{(j-1)/j}, \|a'\|_{L^\infty(T)}^{1/j}\right\}.$$

The function in (728.2) is the absolute value of the derivative of any continuous root, almost everywhere.

Second, the polynomial-root theorem [1, Theorem 1] gives a uniform strong L^q derivative estimate in degree d whenever $q < d/(d-1)$, in terms of the $C^{d-1,1}$ coefficient norms and the interval length. We will apply this estimate only with $d < n$. Degree one is immediate. We also use the already known absolute continuity of degree-n roots, which does not assume the endpoint conclusion.

Lemma 1.2. *If $f \in C^{m-1,1}(I)$ and $\ell = |I| > 0$, then*

$$\ell^s \|f^{(s)}\|_{L^\infty(I)} \le C_m \left(\|f\|_{L^\infty(I)} + \ell^m \,\mathrm{Lip}_I(f^{(m-1)})\right), \qquad 0 \le s \le m-1. \tag{728.3}$$

Proof. Scale I to a unit interval and subtract the Taylor polynomial of degree $m-1$ at its midpoint. Lipschitz continuity of the top derivative bounds all derivatives of the remainder through order $m-1$ by $C_m \,\mathrm{Lip}(f^{(m-1)})$. On the finite-dimensional space of polynomials of degree at most $m-1$, every derivative supremum is bounded by a constant times the polynomial's supremum. The latter is at most $\|f\|_\infty + C_m \,\mathrm{Lip}(f^{(m-1)})$. Rescaling proves (728.3). The argument applies to complex-valued functions as well. □

1.2 Reduction and the stopping intervals

Put

$$R = \max_{1\le j\le n} \|a_j\|_{C^{n-1,1}([0,1])}^{1/j}.$$

The Tschirnhausen substitution $z = w - a_1(t)/n$ removes the coefficient of w^{n-1}. Its new coefficients $\widetilde{a}_j$, $2 \le j \le n$, are polynomials of weighted degree j in the original coefficients, where a_i has weight i. Leibniz's rule therefore gives

$$\|\widetilde{a}_j\|_{C^{n-1,1}([0,1])} \le C_n R^j.$$

Extend these functions to $T = (-2, 3)$ by a bounded $C^{n-1,1}$ extension operator. Endpoint Taylor extensions already provide such an operator on this fixed larger interval. After changing C_n, the same bounds hold on T. Since $\widetilde{\lambda} = \lambda + a_1/n$, it suffices to prove the estimate for the transformed root and then add the bounded derivative a_1'/n. Suppress tildes for the moment, so that $a_1 = 0$.

Define

$$A_0 = \max_{2\le j\le n} \|a_j\|_{C^{n-1,1}(T)}^{1/j}, \qquad A(t) = \max_{2\le j\le n} |a_j(t)|^{1/j}.$$

If $A_0 = 0$, the transformed root is zero. Hence assume $A_0 > 0$. Let g_j be the function in (728.2) for a_j, and set

$$h(t) = 2A_0 + \sum_{j=2}^{n} g_j(t) \quad (t \in T), \qquad h(t) = 0 \quad (t \notin T). \tag{728.4}$$

For $j < n$, the $C^{j-1,1}$ norm needed in (728.2) is controlled by the $C^{n-1,1}$ norm; for $j = n$ it is exactly the available regularity. Since T has fixed finite length and $j/(j-1) \ge p$, the radical estimate and the finite-sum inequality for weak norms give

$$h \in L^{p,\infty}(\mathbb{R}), \qquad \|h\|_{p,\infty;\mathbb{R}} \le C_n A_0. \tag{728.5}$$

In particular h is locally integrable, since $p > 1$.

Fix a sufficiently small constant $\eta > 0$, depending only on n; its required smallness will be specified by the splitting construction below. Take $\eta < 1/3$. For each

$$t \in E = \{t \in (0,1) : A(t) > 0\},$$

choose the centered open interval I_t satisfying

$$\int_{I_t} h(s)\,ds = \eta A(t). \tag{728.6}$$

The radius exists uniquely: the integral is continuous and strictly increasing while the interval lies in T, and $h \geq 2A_0$ there. If $\ell_t = |I_t|$, then

$$\ell_t \leq \frac{\eta A(t)}{2A_0} \leq \frac{\eta}{2}. \tag{728.7}$$

Thus $I_t \subset T$, and $I_t \cap (0,1)$ has length between $\ell_t/2$ and ℓ_t.

Choose continuous jth roots of a_j on I_t. Absolute continuity and (728.6) imply that their variation from the center is at most $\eta A(t)$. With $A = A(t)$ this gives

$$\sup_{I_t} |a_j| \leq (1+\eta)^j A^j, \qquad \sup_{s\in I_t} \frac{|a_j(s) - a_j(t)|}{A^j} \leq j(1+\eta)^{j-1}\eta. \tag{728.8}$$

Let $L_j = \operatorname{Lip}_T(a_j^{(n-1)}) \leq A_0^j$. From (728.7),

$$\frac{L_j \ell_t^n}{A^j} \leq \left(\frac{\eta}{2}\right)^n \left(\frac{A}{A_0}\right)^{n-j} \leq \left(\frac{\eta}{2}\right)^n. \tag{728.9}$$

Apply Lemma 1.2 with $m = n$. Equations (728.8)–(728.9) show that all the functions

$$s \longmapsto \frac{a_j(t + \ell_t s)}{A^j}, \qquad -\tfrac{1}{2} \leq s \leq \tfrac{1}{2}, \tag{728.10}$$

have uniformly bounded $C^{n-1,1}$ norms, with constants depending only on n.

1.3 Uniform splitting and the gain in the local exponent

Consider the compact set

$$K = \left\{ (c_2, \ldots, c_n) \in \mathbb{C}^{n-1} : \max_{2\leq j\leq n} |c_j|^{1/j} = 1 \right\}.$$

For every $c \in K$, the polynomial $z^n + \sum_{j=2}^n c_j z^{n-j}$ has at least two distinct roots. Indeed, if all roots were equal, the missing z^{n-1} coefficient would force their common value to be zero, contradicting $c \in K$.

Group the distinct roots into two nonempty groups and form the corresponding monic coprime factors B and C. The derivative of the multiplication map on their coefficient vectors is

$$(\dot B, \dot C) \longmapsto C\dot B + B\dot C, \qquad \deg \dot B < \deg B, \quad \deg \dot C < \deg C.$$

This map is injective: coprimality implies that B divides $\dot B$ when $C\dot B + B\dot C = 0$, and the degree bound forces $\dot B = 0$, then $\dot C = 0$. Equal dimensions make it invertible. The analytic implicit function theorem therefore gives an analytic factorization chart near c, in which both factors remain coprime and have degrees less than n.

Choose nested neighborhoods for these charts, with the closures of the inner neighborhoods contained in the outer ones. Compactness of K provides a finite inner cover, a positive uniform margin, and uniform bounds on the finitely many derivatives of the factor-coefficient maps needed below. At the center of I_t, the fixed-scale coefficient vector

$$\left(\frac{a_2(t)}{A^2}, \ldots, \frac{a_n(t)}{A^n}\right)$$

belongs to K. Choose η small enough that (728.8) keeps its entire path on I_t inside a single outer chart, within the chosen margin. Together with (728.10), the chain rule then gives uniformly bounded $C^{n-1,1}$ norms for the coefficients of both normalized factors on the unit rescaled interval.

Fix

$$\frac{n}{n-1} < q < \frac{n-1}{n-2}. \tag{728.11}$$

The root under consideration belongs throughout the connected interval $I_t \cap (0,1)$ to one of these two factors. It cannot switch factors: their root sets are disjoint at every parameter, so the choice of factor is locally constant. If this factor has degree $d < n$, then $q < d/(d-1)$ when $d \geq 2$; degree one is elementary. Apply the established strong subcritical theorem in degree d to

$$\widehat{\lambda}(s) = \frac{\lambda(t + \ell_t s)}{A}.$$

The rescaled parameter interval has length between $1/2$ and 1, and its normalized factor coefficients have bounded $C^{d-1,1}$ norms. Consequently

$$\int_{I_t \cap (0,1)} |\lambda'(s)|^q \, ds \leq C_{n,q} A(t)^q |I_t|^{1-q}. \tag{728.12}$$

This is the essential gain: the local exponent is strictly greater than p, because the polynomial has split into smaller degrees. No above-endpoint estimate in degree n has been assumed.

1.4 Patching by disjoint cores and a maximal function

Let J_t be the concentric interval with one fifth the length of I_t. The $5r$ covering lemma selects a countable pairwise disjoint family $J_i = J_{t_i}$ such that

$$E \subset \bigcup_i 5J_i = \bigcup_i I_{t_i}.$$

Write $I_i = I_{t_i}$ and $A_i = A(t_i)$. We do not need bounded overlap of the larger intervals I_i.

Let $\mathcal{M}$ be the uncentered Hardy–Littlewood maximal operator on $\mathbb{R}$. By Chebyshev's inequality and (728.12), for $u > 0$,

$$\bigl|\{s \in I_i \cap (0,1) : |\lambda'(s)| > u\}\bigr| \leq |I_i| \min\left\{1, C_{n,q}\left(\frac{A_i}{u|I_i|}\right)^q\right\}.$$

Equation (728.6) identifies $A_i/|I_i| = \eta^{-1}|I_i|^{-1}\int_{I_i} h$. For every $x \in J_i \subset I_i$, this average is at most $\mathcal{M}h(x)$. Since $|I_i| = 5|J_i|$, summing over the disjoint cores therefore gives

$$\bigl|\{s \in E : |\lambda'(s)| > u\}\bigr| \leq 5 \int_{\mathbb{R}} \min\{1, C_{n,q}(\mathcal{M}h(x)/u)^q\} \, dx. \tag{728.13}$$

The known absolute continuity of the root implies that its derivative vanishes almost everywhere on $\{A = 0\}$, where the transformed root is zero. Thus the left side of (728.13) can be replaced by the distribution function on all of $(0,1)$.

For completeness, $\mathcal{M}$ is bounded on $L^{p,\infty}$ when $p > 1$. One can see this directly by splitting h at height $u/2$: the maximal function of the lower part is at most $u/2$, while the weak $(1,1)$ estimate applied to the upper part and integration of its distribution function give

$$|\{\mathcal{M}h > u\}| \leq C_p u^{-p} \|h\|_{p,\infty}^p.$$

Put $H = \|\mathcal{M}h\|_{p,\infty}$. For any fixed $D > 0$, layer-cake integration and $q > p$ yield

$$\begin{aligned} \int_{\mathbb{R}} \min\{1, (D\mathcal{M}h/u)^q\} \, dx &= q(D/u)^q \int_0^{u/D} v^{q-1} |\{\mathcal{M}h > v\}| \, dv \\ &\leq \frac{q}{q-p}(DH/u)^p. \end{aligned} \tag{728.14}$$

Combining (728.5), (728.13), and (728.14), with a fixed choice of q in (728.11), proves

$$\|\widetilde{\lambda}'\|_{p,\infty;(0,1)} \leq C_n A_0 \leq C_n R.$$

Restoring $\lambda = \widetilde{\lambda} - a_1/n$ and using the finite-sum inequality for weak norms proves (728.1).

1.5 An explicit estimate on any bounded interval

If $I = (a, b)$ has length ℓ, define

$$N_{j,I} = \max\left\{ \max_{0\le s\le n-1} \ell^s \|a_j^{(s)}\|_{L^\infty(I)}, \ell^n \operatorname{Lip}_I(a_j^{(n-1)}) \right\}.$$

Rescaling Theorem 1.1 gives

$$\boxed{\|\lambda'\|_{n/(n-1),\infty;I} \le C_n \ell^{-1/n} \max_{1\le j\le n} N_{j,I}^{1/j}.} \tag{728.15}$$

The dependence on constant coefficient parts is not asserted to be optimal. The estimate is finite under exactly the stated hypotheses and has the correct homogeneity under scaling the root and the parameter. The example $z^n - t$ still excludes ordinary strong endpoint integrability; there is no conflict with that sharp obstruction.

References

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